

Simulation of an Ar/O₂/Cl₂ magnetized direct current discharge

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Abstract

A homemade 2D hybrid model (coupling particle in cell model with fluid model) is used to investigate the effect of the reaction chamber radius, the mixing ratio and pressure on the plasma properties in an Ar/O₂/Cl₂ magnetized DC (direct current) discharge. The results show that the main products are Ar^{**}, Ar^{*}, e, Ar₂⁺, Cl, Cl₂⁺, O(³P) and O₂⁺. The densities of oxygen species are slightly lower than those of Cl₂ species. If the magnetic field strength of the central magnetic pole is slightly weaker than those of the edge magnetic poles, it will improve the uniformity of the plasma species at the film basal position. Even the mixing ratio of Ar/O₂/Cl₂ is varied from 9:0.2:1 to 9:1:0.2, the main particles are hardly changed, but the concentrations of Cl₂⁺ ions and Cl atoms significantly decrease. The main ion densities at 3 mTorr appear to be lower, but more inhomogeneous than those at 6 mTorr.

Keywords: magnetized direct current discharge, oxygen, argon, chlorine, hybrid model

(Some figures may appear in colour only in the online journal)

1. Introduction

Magnetized DC devices have been widely used for thin-film deposition and sputter etching. However, the strong non-uniform magnetic field at the cathode, which significantly distort the profiles of particles and electron temperatures, and then decrease the uniformity and firmness of the film. On the other hand, in order to meet the needs of all kinds of work-piece evaporation, the increasing of the reaction cavity size is an inevitable trend. So the optimization of the magnetic field configuration becomes increasingly prominent. Take a typical electronegative plasma (Ar/O₂/Cl₂) discharge as an example. Yoo *et al* [1] found that the etch slop of Pt increased up by adding O₂ to Ar/Cl₂. Chung *et al* [2] investigated the dry etch behavior of Pt films using a TiO₂ hard mask with an O₂/Cl₂/Ar gas chemistry in an inductively coupled plasma. Smith *et al* [3] demonstrated etching selectivity between GaN and AlN. Lee *et al* [4] examined the selective etch characteristics of GaN,

Al_xGa_{1-x}N and In_xGa_{1-x}N using Cl₂/Ar/O₂ as the etching gas. Hsu *et al* [5] compared the models and experiments for Ar, Ar/O₂ and Ar/O₂/Cl₂ inductively coupled plasma. Tinck *et al* [6] calculated the etching behavior of Cl₂/O₂/Ar inductively coupled plasma on a Si substrate. In a word, the mixing ratio and pressure also significantly influenced the etch selectivity of film [2].

In the present work, a homemade two-dimensional (2D) planar hybrid model (Coupling PIC (Particle in cell) model with fluid model) is used to investigate the effect of the reaction chamber radius, gas mixing ratio and pressure on the properties of the Ar/O₂/Cl₂ discharge in the magnetized DC reactor. To check the efficiency and the accuracy, we compare this calculated results of the Ar/O₂/Cl₂ discharge with the available experimental data.

2. Model description

In the hybrid model electrons are split into high-energy group with electron energies higher than E_{thr} and low-energy group

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with electron energies less than E_{thr} . E_{thr} is normally the threshold energy for the lowest energetic interaction of consideration. E_{thr} is set to 7.8 eV. The fast electrons are firstly treated by the Monte Carlo code. For each electron the Newton equation of motion with a Lorentz force term is solved in every time step. For the first run a guess is needed for electric field E . Their trajectories are traced from the moment an electron is created until it disappears. The latter could be due to transfer to the slow (bulk) electron group or absorption by the electrodes.

Collisions between fast electrons and neutrals are treated with Monte Carlo technique [7]. The probability and the type of collision are determined using a single random number, which is compared to the normalized probability P_k of the k th collision event [8].

$$P_k(\varepsilon) = N\sigma_k(\varepsilon)\sqrt{\frac{2\varepsilon}{m_e}}\Delta t \quad (1)$$

where N is the densities of the background gases, σ_k is the cross-section for the k th event, m_e is the electron mass and ε is the electron energy. The time step Δt for Monte Carlo code is normally around 10^{-12} s.

The deflection angle in all types of scattering is [8]:

$$\chi = \arccos\left[1 - 2\frac{R}{1 + 8\varepsilon(1 - R)}\right] \quad (2)$$

where R is a uniformly distributed random number in the interval $[0,1]$, $\varepsilon = E_e/E_0$, E_e is the energy of the incident electron and E_0 is Bohr energy.

Once the fast electron density and the creation rates for the bulk electrons and ions are known, the fluid code is executed to produce the slow electron and ion density distribution, electric field profile and ion flux at the cathode.

Electrons are highly restricted in their movement by the magnetic field. The electron flux is expressed as [8]:

$$\Gamma_r^{e-} = -\frac{\mu_{B=0}^{e-}}{1 + \frac{\omega_r^2}{\nu^2}}n_e E_r - \frac{D_{B=0}^{e-}}{1 + \frac{\omega_r^2}{\nu_i^2}}\frac{\partial n_e}{\partial r} \quad (3)$$

$$\Gamma_z^{e-} = -\frac{\mu_{B=0}^{e-}}{1 + \frac{\omega_z^2}{\nu^2}}n_e E_z - \frac{D_{B=0}^{e-}}{1 + \frac{\omega_z^2}{\nu_i^2}}\frac{\partial n_e}{\partial z} \quad (4)$$

where ν and ω are the mean collision frequency and the gyro frequency, respectively. μ and D are the mobility and diffusion of electrons in the lack of magnetic field, respectively. n_e is the electron density. E is the electric field.

The ion trajectories are not significantly affected in most of the sputtering systems. In the drift-diffusion approximation of the Boltzmann equation, their flux is given by [8]:

$$\Gamma_r^i = \mu_i n_i E_r - D_i \frac{\partial n_i}{\partial r} \quad (5)$$

Table 1. Overview of the plasma species included in the Ar/O₂/Cl₂ discharge model, besides the electrons.

Ground state neutrals	Excited state neutrals	Ions
Ar	Ar*, Ar**, Ar(4p) ^a	Ar ⁺
	Ar ₂ *	Ar ₂ ⁺
O ₂ (X)	O ₂ (a), O ₂ (b), O ₂ (A)	O ₂ ⁺ , O ₂ ⁻
O(³ P)	O(¹ D)	O ⁺ , O ⁻
O ₃		O ₃ ⁻
		O ₄ ^{+b}
Cl ₂ , Cl, ClO	Cl*	Cl ₂ ⁺ , Cl ⁺ , Cl ⁻ , ClO ⁺

^a From [9].

^b From [10].

$$\Gamma_z^i = \mu_i n_i E_z - D_i \frac{\partial n_i}{\partial z} \quad (6)$$

where μ and D are the mobility and the diffusion coefficient of ions, respectively. n_i is the ion density.

For both species the continuity equation of the magnetized DC device is fulfilled [8]:

$$\frac{\partial n}{\partial t} + \nabla \cdot \Gamma = S \quad (7)$$

where S is the net creation rates of electrons and ions.

29 different species are adopted to describe the chemical reactions in the Ar/O₂/Cl₂ mixture (see table 1). In total, 264 reactions and the corresponding rate coefficients (or cross sections) for the Ar/O₂/Cl₂ plasma chemistry are used. The plasma reactions are included in the Ar/O₂/Cl₂ discharge model, besides the literature [5, 6, 9, 11–15] (see table 2). Rate coefficients in table 2 are cm³·s⁻¹ for 2-body reactions and cm⁶·s⁻¹ for 3-body reactions. For radiation reactions, units are s⁻¹. The cross sections of argon are calculated by the formula from XOOPIC sources [26]. The cross sections of oxygen are obtained from [29]. The reaction coefficients of different electron processes are calculated from the cross sections by the formula in [30].

The energy equation is solved for electrons [31]:

$$\frac{\partial}{\partial t}\left(\frac{3}{2}k_B T_e n_e\right) + \frac{\partial q_e}{\partial r} + e j_e E + p_{\text{coll}} = 0 \quad (8)$$

where k_B is the Boltzmann constant, T_e is the electron temperature, q_e is the electron heat flux, e is the electron charge, P is the collision energy loss with the background gas. Threshold energies for Ar, O₂, Cl₂ excitation are adopted from the [5, 12, 30], respectively.

The electrical field is obtained from the Poisson equation [32]:

$$\nabla^2 V = -\frac{e}{\varepsilon_0}(n_+ - n_- - n_e) \quad (9)$$

where ε_0 is the vacuum permittivity, n_+ and n_- are the positive and negative ion densities, respectively. n_e is the total

Table 2. Reactions are taken into account in our model, besides the literature [5, 6, 9, 11–15].

Reactions	Rate coefficients	References
$e + \text{Ar} \rightarrow \text{Ar} + e$		[16] ^a
$e + \text{Ar} \rightarrow \text{Ar}^* + e$	$3.6 \times 10^{-7} \exp(-57.7/T_e)$	[17] ^b
$e + \text{Ar} \rightarrow \text{Ar}^{**} + e$	$2.1 \times 10^{-5} \exp(-74.7/T_e)$	[17] ^b
$e + \text{Ar}^{**} \rightarrow \text{Ar}^+ + 2e$	$2.1 \times 10^{-7} \exp(-4.95/T_e)$	[18]
$e + \text{Ar}^+ \rightarrow \text{Ar}$	$4.0 \times 10^{-13} T_e^{-0.5}$	[19] ^b
$e + \text{Ar}^+ \rightarrow \text{Ar}^*$	$4.0 \times 10^{-13} T_e^{-0.5}$	[20]
$e + \text{Ar}_2^+ \rightarrow 2\text{Ar}$	$8.5 \times 10^{-7} (T_e/300)^{-0.67} (T_g/300)^{-0.58}$	[21]
$e + \text{Ar}_2^+ \rightarrow \text{Ar}^* + \text{Ar}$	$7 \times 10^{-7} (300/T_e)^{0.5}$	[16]
$\text{Ar}^* \rightarrow \text{Ar}$	3.2×10^7	[15]
$\text{Ar}^* + \text{Ar}^* \rightarrow \text{Ar}_2^+ + e$	6×10^{-10}	[22]
$\text{Ar}^* + \text{Ar}^{**} \rightarrow \text{Ar}_2^+ + e$	2.1×10^{-9}	[23]
$\text{Ar}^{**} + \text{Ar} \rightarrow 2\text{Ar}$	2.3×10^{-15}	[24]
$\text{Ar}^{**} + \text{Ar} \rightarrow \text{Ar}_2^+ + e$	2×10^{-9}	[21]
$\text{Ar}^{**} + \text{Ar}^* \rightarrow \text{Ar}_2^+ + e$	5.7×10^{-10}	[21]
$\text{Ar}^{**} + \text{O}_2 \rightarrow \text{Ar} + \text{O}_2$	2.5×10^{-10}	[10]
$e + \text{O}_2 \rightarrow e + 2\text{O}(^3\text{P})$	$7.1 \times 10^{-9} \exp(-8.6/T_e)$	[25]
$e + \text{O}_2 \rightarrow e + \text{O}(^3\text{P}) + \text{O}(^1\text{D})$	$1.8 \times 10^{-7} \exp(-18.33/T_e)$	[25]
$e + \text{O}_2 \rightarrow e + \text{O}_2(\text{a})$	$1.7 \times 10^{-9} \exp(-3.1/T_e)$	[25]
$\text{Ar}^+ + \text{Ar} \rightarrow \text{Ar}^+ + \text{Ar}^*$		[26] ^a
$\text{Ar}^+ + \text{Ar} \rightarrow \text{Ar}^+ + \text{Ar}^{**}$		[26] ^a
$\text{Ar}^+ + \text{Ar} \rightarrow 2\text{Ar}^+ + e$		[26] ^a
$e + \text{O}_4^+ \rightarrow 2\text{O}_2$	$2 \times 10^{-6} (300/T_e^{-0.5})$	[10]
$e + \text{O}_2^+ \text{Ar} \rightarrow \text{O}_2^- + \text{Ar}$	3×10^{-32}	[10]
$e + 2\text{O}_2 \rightarrow \text{O}_2^- + \text{O}_2$	$2.26 \times 10^{-30} (300/T_g^{0.5})$	[10]
$\text{Ar}^* + 2\text{Ar} \rightarrow 3\text{Ar}$	1.4×10^{-32}	[22]
$\text{Ar}^+ + 2\text{Ar} \rightarrow \text{Ar}_2^+ + \text{Ar}$	$2.5 \times 10^{-31} \times (300/T_e)^{1.5}$	[16]
$\text{O}_2^+ + \text{O}_2^+ \text{Ar} \rightarrow \text{O}_4^+ + \text{Ar}$	8.0×10^{-30}	[10]
$\text{O}_2^+ + 2\text{O}_2 \rightarrow \text{O}_4^+ + \text{O}_2$	2.5×10^{-30}	[10]
$\text{O}_4^+ + \text{O}_2^- + \text{Ar} \rightarrow 3\text{O}_2^+ \text{Ar}$	2×10^{-25}	[10]
$\text{O}_4^+ + \text{O}_2^- + \text{O}_2 \rightarrow 4\text{O}_2$	2×10^{-25}	[10]
$e + \text{Cl}_2 \rightarrow e + \text{Cl}_2$		[27] ^a
$e + \text{Cl}_2 \rightarrow \text{Cl}^- + \text{Cl}$	$7.2834 \times 10^{-10} T_e^{-0.7162}$	[28] ^c
$e + \text{Cl}_2^+ \rightarrow 2\text{Cl}$	$9 \times 10^{-7} T_e^{-0.6}$	[28]
$e + \text{Cl}^* \rightarrow \text{Cl}^+ + 2e$	$9.29 \times 10^{-9} T_e^{0.265} \exp(-47436.0/T_e)$	[28]
$e + \text{Cl} \rightarrow \text{Cl}^+ + 2e$	$1.0876 \times 10^{-10} T_e^{2.3785}$	[28] ^c
$\text{O} + \text{Cl}^- \rightarrow \text{ClO} + e$	1×10^{-11}	[27] ^a

^a Rate coefficient is calculated by using cross-section data from the indicated reference.

^b Estimated by the indicated reference.

^c Estimated by using figure 5.5 on page 105 from [28].

density of n_{fast}^e and n_{slow}^e , n_{fast}^e is taken as input from the MC code.

The electric field intensity E is calculated from [8]:

$$E = -\nabla V. \quad (10)$$

The boundary conditions are as follows: at the cathode, $\Gamma_{e,B} = -\gamma \Gamma_{i,B}$ [32, 33]. Where γ is the secondary electron emission coefficient, it is set at a constant value of 0.12 for the

electrode surfaces [34]. At the walls, $\Phi = 0$ (ground), at the cathode, $\Phi = -V_{\text{applied}} = -200\text{V}$ [35, 36]. At the walls, T_e is assumed to be 0.5 eV and $\nabla T_e = 0$ [37].

The electron mobility and the electron diffusion coefficient are given by equations (6) and (7) in [38]. The ion mobility is calculated using the mobility expression from [39]. The polarizability is from [40]. The ion diffusion coefficient is calculated from the mobility using the Einstein relation from [41].

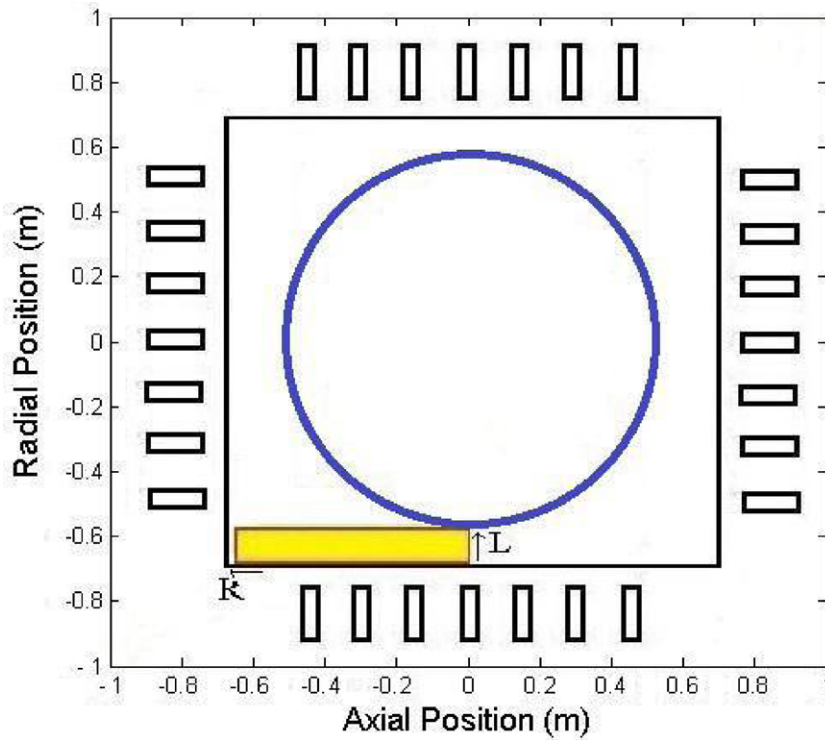


Figure 1. The scheme of the cylindrical symmetric magnetized DC apparatus.

For the neutrals, diffusion coefficients are calculated by the Chapman–Enskog equation using the characteristic Lennard–Jones parameters (σ_j (Å) and ε_j (K)): for Ar: 3.330; 136.500; [42] for O_2 : 3.458; 107.4; [43] for Cl_2 : 3.402; 160.86 [44]. The diffusion coefficient D_j of the species j in the entire gas mixture is obtained from the different D_{ij} values [41]. The ion temperature is given by [45].

The electric field profile and ion flux are given back as input to the Monte Carlo code and the whole procedure is repeated until a steady state is obtained [35, 36].

3. Results and analysis

The typical structure of the cylindrical symmetric magnetized DC apparatus adopted is similar to [46, 47]. Discharge deal with a titanium cathode and the anode is grounded [48]. At our experiment center, we extended the radius of the reactor to evaporate larger workpieces and measured the magnetic field near the magnetic poles. The vertical scheme of the cylindrical symmetric magnetized DC apparatus investigated in the present study is plotted in figure 1. The radius of the reaction chamber is 68 cm. The width of each magnetic pole is set to 5 cm and the width between the magnetic poles is 10 cm. The blue region is a rotary tray (as the substrate) to uniform coat the workpieces. The plasma parameters in the yellow region in figure 1 are considered.

Red line region in figure 1 is chosen bi-dimensional (R , L). The origin of the R -axis and L -axis is the center of the magnetic poles. As an initial condition, the background

Table 3. The input parameters of the theoretical model.

Setting	Value
Voltage	−200 V ^a
Maximal B	0.11 T ^{b,c}
Reactor pressure	5 mTorr ^a
Gas composition Ar/ O_2 / Cl_2 (vol.%)	9/0.2/1 ^d
Electrode distance	10 cm ^b
Wall temperature	323 K ^a
Secondary electron emission coefficient γ	0.12 ^e

^a From [49].

^b From [50].

^c From [47].

^d From [51].

^e From [34].

gas has constant density [35] and flow of background gases are ignored. Supposed all the initial neutral and ion density are 10^{10} cm^{-3} and 10^{11} cm^{-3} , respectively. The electron density is the sum of all the ion densities. The initial fast electron density is taken as zero [7]. The initial electron temperature T_e is 0.5 eV. The simulations are carried out until all major discharge parameters converge to the steady-state values [35]. The detailed input parameters for the theoretical model are listed in table 3.

3.1. Reaction chamber radius

To investigate the effect of the reaction chamber radius on the densities of the main plasma species and electron temperature,

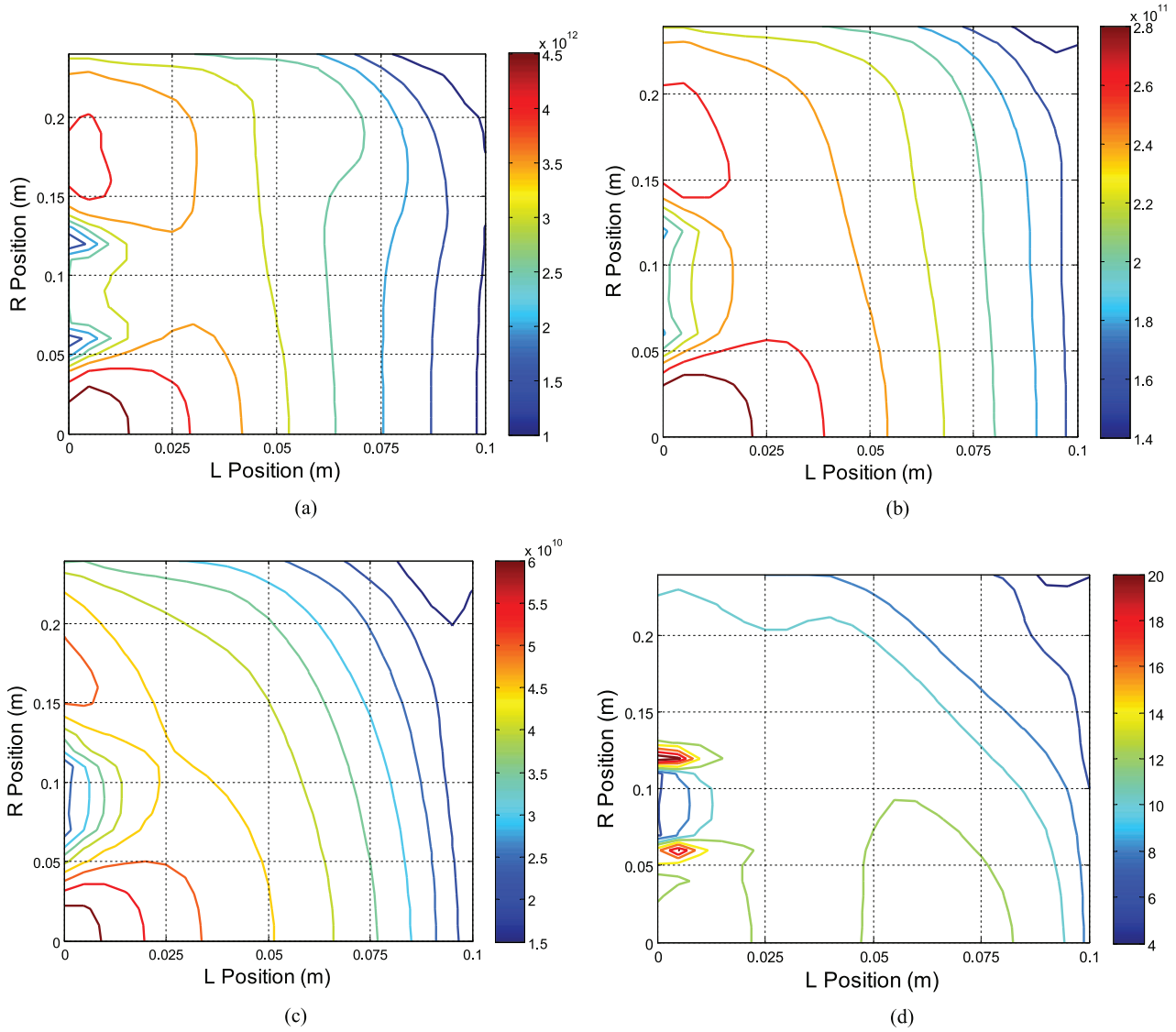


Figure 2. The calculated profiles as the reaction chamber radius are 24 cm. (a) n_e , (b) n_{Cl} , (c) n_O , (d) T_e .

the former is changed from 24 cm, 46 cm to 68 cm. The width and distance of magnetic poles are set to constant. The simulations are carried out with all other plasma parameters are kept constant (see table 3). The calculated potential and electric field profiles are similar to those in the [24]. The main particles are Ar^{**} , Ar^* , e , Ar_2^+ , Cl , Cl_2^+ , $O(^3P)$ and O_2^+ besides the background gases. The result is agreement with that in the [6]. Due to the distributions of the main species are similar, to control the quality of the deposited film, only the electron density, Cl density, O density and electron temperature with the reaction chamber radius (24 cm, 46 cm and 68 cm) are given in figures 2–4.

Comparing figures 2–4, it can find that the main plasma particles gather to the magnetic poles and become more uniform at 10 cm in front of magnetic poles. All of the electron densities are very well localized at the magnet poles and have the maximum of $4.75 \times 10^{12} \text{ cm}^{-3}$, $1.31 \times 10^{12} \text{ cm}^{-3}$ and $3.51 \times 10^{12} \text{ cm}^{-3}$, respectively. Goeckner *et al*

[46] found that the electron density was the highest near the center of the electron trap and it became more uniform near the anode. Lin *et al* [52] illustrated the electron motion in the electromagnetic field. Fast electrons could be transferred to the bulk electrons or be absorbed by the electrodes [8]. In the bulk region, fast electrons are mainly transferred by exciting the background gases. ($e + Ar \rightarrow Ar(4p) + e$ [9], $e + O_2 \rightarrow e + O(^3P) + O(^1D)$ [25], $e + O_2 \rightarrow e + 2O(^3P)$ [25], $e + Cl_2 \rightarrow e + 2Cl$ [5], $e + Cl_2 \rightarrow e + Cl_2(^1\Sigma_u)$ [5], $e + Ar^* \rightarrow Ar(4p) + e$ [9] The reaction coefficients of $e + Ar \rightarrow Ar^* + e$ [17] and $e + Ar \rightarrow Ar^{**} + e$ [17] are three orders of lower than that of $e + Ar \rightarrow Ar(4p) + e$ [9]. The reaction coefficients of electrons colliding with O_2 which transfer O_2 to $O(a)$ [25], $O(A)$ [9] and O_2^+ [13] are one order lower than those of O_2 transfer to $O(^3P)$ and $O(^1D)$. The reaction coefficients of electrons colliding with Cl_2 which transfer Cl_2 to Cl_2^+ [5], $Cl_2(\text{Ryd})$ [5] and $Cl_2(b^3\Pi_u)$ [5] are one order lower than those of Cl_2 transfer

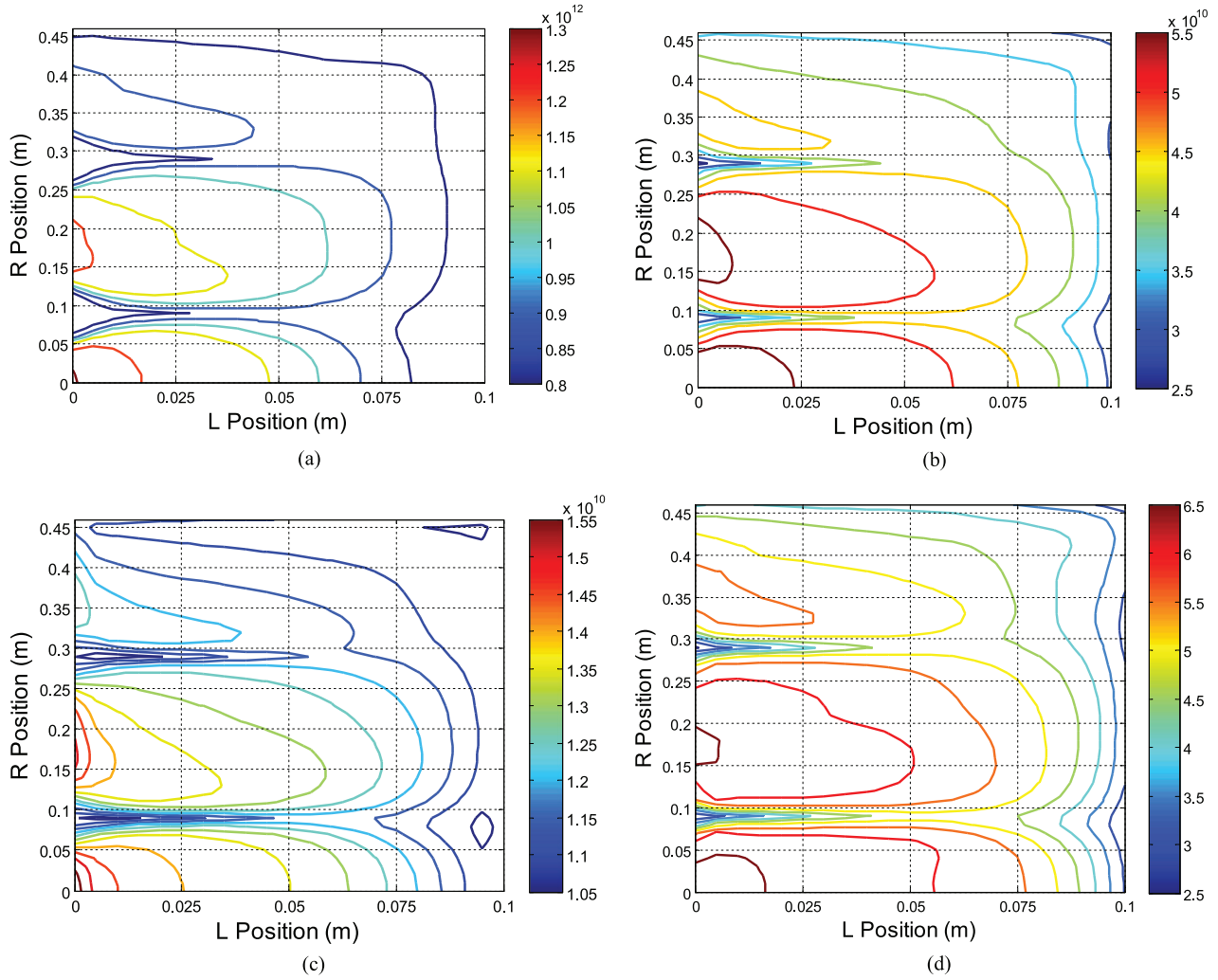


Figure 3. The calculated profiles as the reaction chamber radius are 46 cm. (a) n_e , (b) n_{Cl} , (c) n_O , (d) T_e .

to Cl and $Cl_2(^1\Sigma_u)$. Cl_2 transfers to Cl_2^+ which becomes the main reaction between electrons and Cl_2 . In short, fast electrons prior accelerated the indirect ionization in the bulk region. In the sheath, the electron excitation reactions are still dominant. The probability and the type of collision [8] and the reaction coefficients of different electron processes [30] change with the cross sections change. The reaction coefficients of $e + Ar \rightarrow Ar(4p) + e$ and $e + O_2 \rightarrow e + O(^3P) + O(^1D)$ are two orders and one order higher than those in the bulk region.

Profiles of the main plasma densities are similar to those of electrons. The ratios of the maximum density between Ar^* and Ar^{**} are 23.07%, 41.02% and 29.6%, respectively. O_2 added to Ar which will decrease the meta-stable density [53]. Additions of Cl_2 were found to significantly modify the spatial distribution of meta-stable densities [53]. The dominant ion in the Ar/ O_2 / Cl_2 discharge is Ar_2^+ not Ar^+ . The maximums of Ar_2^+ densities are $3.92 \times 10^{12} \text{ cm}^{-3}$, $6.65 \times 10^{11} \text{ cm}^{-3}$ and $3.02 \times 10^{12} \text{ cm}^{-3}$, respectively. The densities of Ar^+ are relatively small and they are only 2.8%, 15.0% and 3.7% of those of Ar_2^+ . The behavior of Ar^+ ions is influenced by non-resonant exothermic charge exchange with O_2 and O [49, 54]. The

presence of negative ions can act as an additional loss channel for Ar^+ through ion-ion recombination [29]. The maximums of Cl_2^+ densities are $2.79 \times 10^{11} \text{ cm}^{-3}$, $1.26 \times 10^{11} \text{ cm}^{-3}$ and $2.10 \times 10^{11} \text{ cm}^{-3}$, respectively. The densities of oxygen species are significantly lower than others. It's mainly because that oxygen concentration is only one fifth of that of chlorine. For the ionic species, the ionization potential of oxygen is higher than that of chlorine and argon [5, 12, 54]. The maximums of Cl atom density are $2.99 \times 10^{11} \text{ cm}^{-3}$, $5.99 \times 10^{10} \text{ cm}^{-3}$ and $1.81 \times 10^{11} \text{ cm}^{-3}$, respectively. The maximums of O atom density are $6.30 \times 10^{10} \text{ cm}^{-3}$, $1.57 \times 10^{10} \text{ cm}^{-3}$ and $3.53 \times 10^{10} \text{ cm}^{-3}$, respectively.

The nonlinear variation of the plasma density profiles is derived from those of the electron temperatures. Near the magnetic poles, the electron temperatures are significantly high. Electron temperatures near the magnetic poles are about 21.95 eV, 6.77 eV and 12.42 eV but near the cathode they are around 7.42 eV, 3.30 eV and 4.53 eV, respectively. It shows that the important significance of design the magnetic field when the reaction chamber is expanded. From figures 2 and 4 it can be seen that if the asymmetric magnetic poles are adopted which means that the magnetic field strength of the

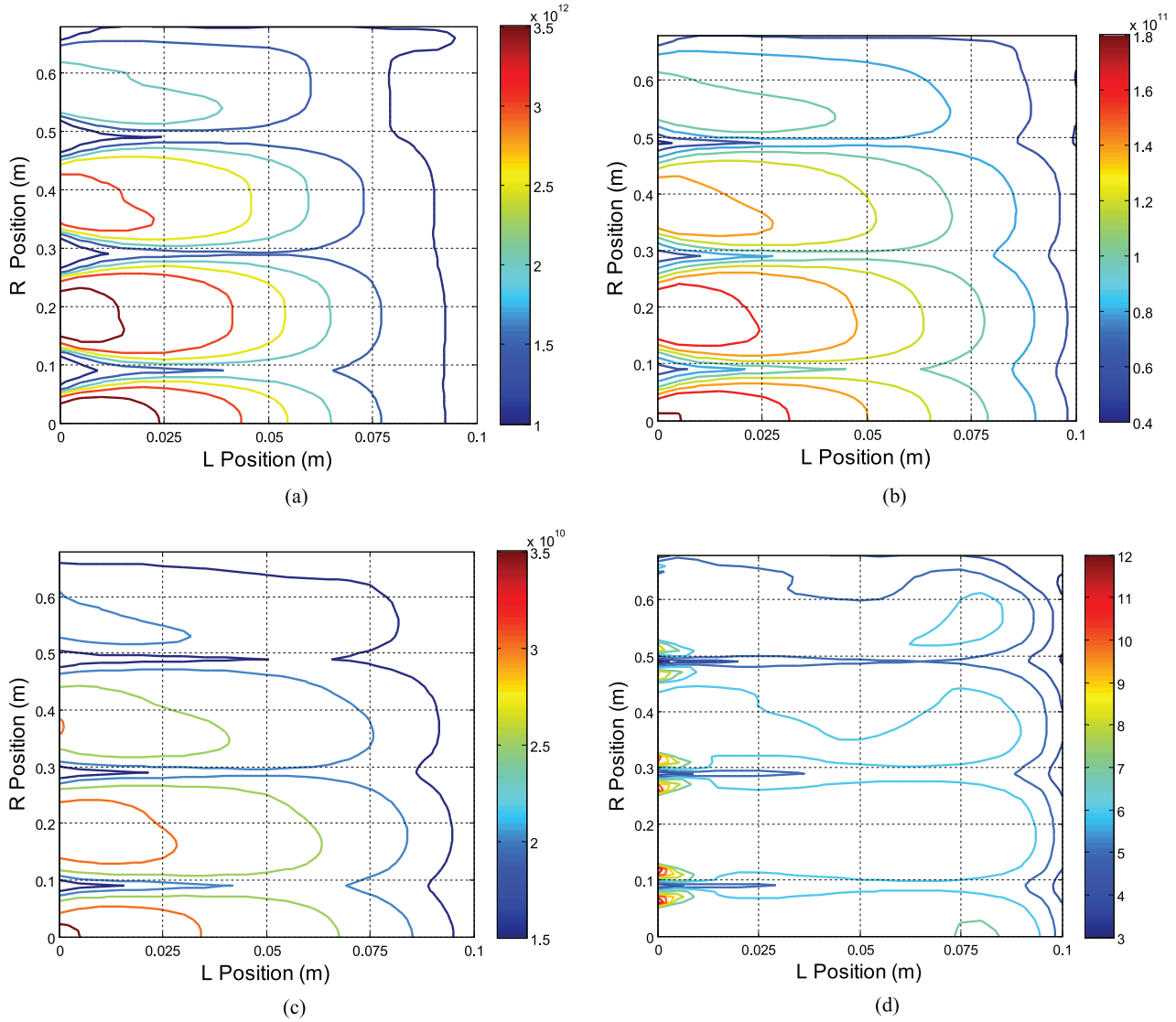


Figure 4. The calculated profiles as the reaction chamber radius are 68 cm. (a) n_e , (b) n_{Cl} , (c) n_O , (d) T_e .

central magnetic pole is slightly weaker than those of the edge magnetic poles, it will improve the uniformity of the plasma density at the film basal position.

3.2. Mixing ratio

To investigate the effect of the mixing ratio of the densities of the main plasma species and electron temperature in Ar/O₂/Cl₂ discharge, according to the experimental parameters [1–4], the mixing ratio is chosen in sequence 9:0.2:1, 9:0.5:1, 9:1:0.5 and 9:1:0.2. Simulations are carried out with all other plasma parameters are kept constant (see table 3). The results of the mixing ratio is 9:0.2:1 have been discussed in the section 1. The ratios of the maximum density between Ar* and Ar** are 28.3%, 28.0% and 29.3%, respectively. The maximums of Ar** density are $1.54 \times 10^{14} \text{ cm}^{-3}$, $1.58 \times 10^{14} \text{ cm}^{-3}$ and $1.62 \times 10^{14} \text{ cm}^{-3}$, respectively. It is because that the excitation threshold energy (11.48 eV [5]) of Cl₂ is slightly lower than that (12.20 eV [15]) of O₂. The maximums of Ar₂⁺ densities are $2.9 \times 10^{12} \text{ cm}^{-3}$, $3.0 \times 10^{12} \text{ cm}^{-3}$ and $3.2 \times 10^{12} \text{ cm}^{-3}$,

respectively. The densities of Ar⁺ are all less than 3.7% of those of Ar₂⁺. Farouk has found that the predicted density of Ar₂⁺ is found to be one order of magnitude higher than that of Ar⁺ [55]. The maximums of Cl₂⁺ density are $2.09 \times 10^{11} \text{ cm}^{-3}$, $1.56 \times 10^{11} \text{ cm}^{-3}$ and $1.20 \times 10^{11} \text{ cm}^{-3}$, respectively. The magnitude of the calculated ion densities (Ar⁺, Cl₂⁺) is an agreement with the figure 13 in the [5]. Additions of O₂ will decrease the relative density of Chlorine, decrease the ion current density, and then decrease the etch rate of PtCl_xO_y from the surface [51].

Only the electron densities, Cl densities, O densities and electron temperatures of the mixing ratio (9:0.2:1 and 9:1:0.2) are shown in figures 4 and 5. The changing trend and magnitude of the calculated radical densities (Cl and O) and ion densities (Ar⁺, Cl₂⁺) are agreement with the figures 12 and 13 in the [5]. As the ratio of Cl₂ decreases, the concentrations of Cl atoms and Cl₂⁺ ions significantly decrease, but the concentrations of electron, Ar**, Ar*, Ar₂⁺ and O atoms slightly increase. Kwon *et al* [51] have pointed out O₂ largely contributed to the removal of Cl-containing compound and finally

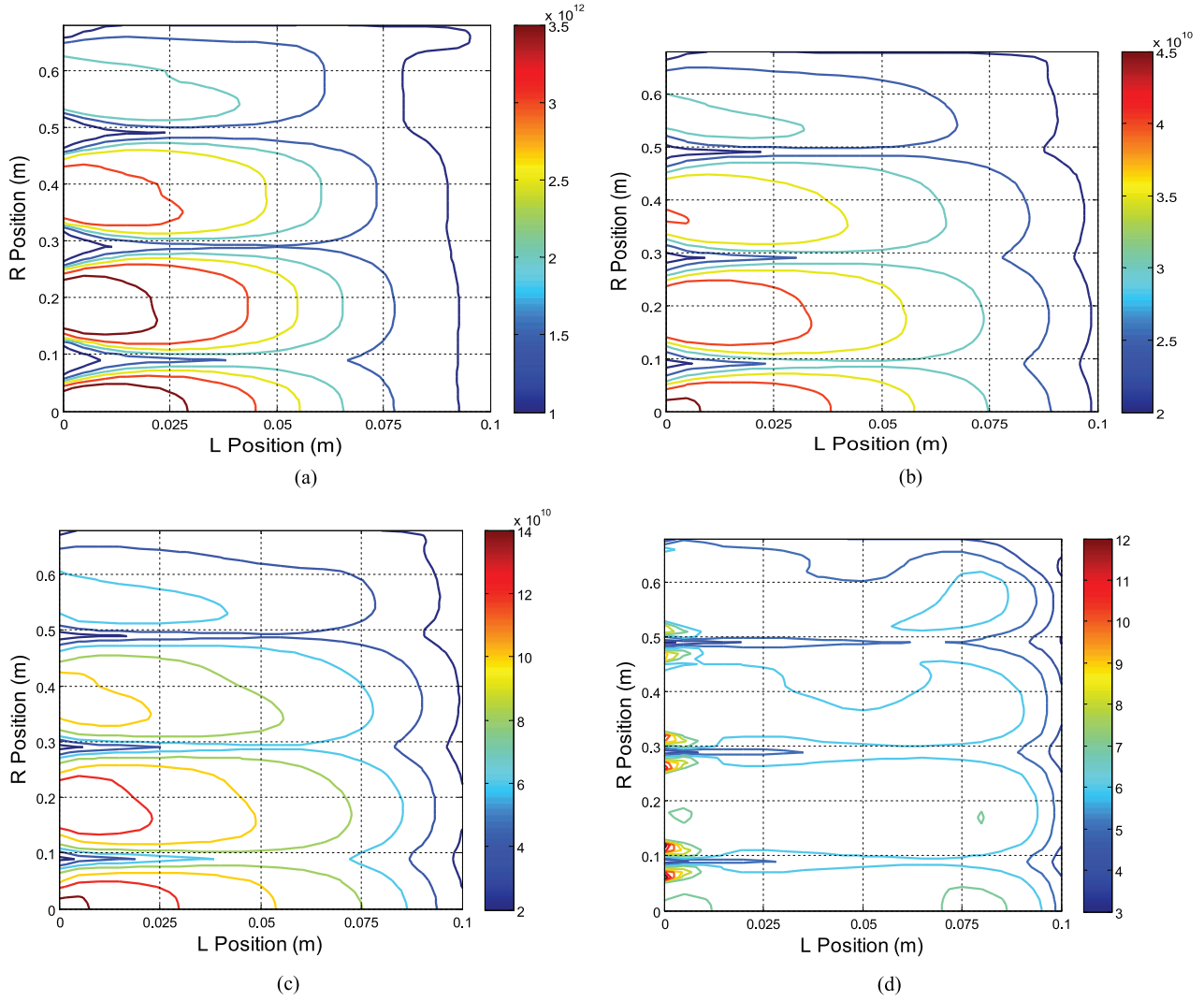


Figure 5. The calculated profiles as the mixing ratio of Ar/O₂/Cl₂ are 9:1:0.2 at 5 mTorr. (a) n_e , (b) n_{Cl} , (c) n_O , (d) T_e .

reduced the atomic specie of Cl on the etched surface. Tinck *et al* [6] found that the oxygen fraction in Cl₂-O₂-Ar inductively coupled plasma slightly increase, the etch rate abruptly drops by simulations and experiments. Comparing figures 4(b) and (c) with figures 5(b) and (c), it can see that to acquire the same concentrations Cl and O, the ratio of Cl₂ should be slightly higher than that of O₂.

The calculated electron temperature maximums slightly increase once the mixing ratio of the Ar/O₂/Cl₂ variation in sequence 9:0.2:1, 9:0.5:1, 9:1:0.5 and 9:1:0.2. In the magnetized regions electron temperatures are about 4.5–4.6 eV, near the cathode they are around 12.2 eV, 12.5 eV and 12.9 eV, respectively. It is because the threshold energy for $e + O_2 \rightarrow O_2^+ + 2e$ (12.20 eV [15]) is higher than that for $e + Cl_2 \rightarrow Cl_2^+ + 2e$ (11.48 eV [5]).

3.3. Pressure

Simulations are done to study the effect of pressure on the Ar/O₂/Cl₂ discharge for pressure changed from 3 mTorr, 4 mTorr to 6 mTorr (the results of 5 mTorr missed because they can be found in the section 1). All other plasma parameters

are kept constant (see table 3). The densities of the plasma species are higher in the reactor chamber when pressure is increased [6]. The ratios of the maximum density between Ar^{*} and Ar^{**} are 32.5%, 26.7% and 30.6%, respectively. The maximums of Ar^{**} densities are $6.6 \times 10^{13} \text{ cm}^{-3}$, $1.2 \times 10^{13} \text{ cm}^{-3}$ and $2.0 \times 10^{14} \text{ cm}^{-3}$, respectively. The calculated dominant ion in the Ar/O₂/Cl₂ mixture is still Ar₂⁺. Farouk has found that the predicted density of Ar₂⁺ is higher than that of Ar⁺ [55]. The maximums of Ar₂⁺ densities are $4.9 \times 10^{11} \text{ cm}^{-3}$, $1.5 \times 10^{12} \text{ cm}^{-3}$ and $4.1 \times 10^{12} \text{ cm}^{-3}$, respectively. The ratios of Cl₂⁺ to the other charged particles increase with the pressure increases. The charge exchange collisions play a more significant role at higher pressure than at lower pressure. The maximums of Cl₂⁺ densities are $1.3 \times 10^{11} \text{ cm}^{-3}$, $1.7 \times 10^{11} \text{ cm}^{-3}$ and $2.3 \times 10^{11} \text{ cm}^{-3}$, respectively.

The electron density, Cl density, O density and electron temperature for the 3 mTorr and 6 mTorr are given in figures 6 and 7, respectively. The electron density is monotonic increases with the pressure increases. It can be explained that the ionization rate increases with the pressure increases. Electron density at 3 mTorr appears more inhomogeneous than that at 6 mTorr. These estimations are in good agreement with the results given

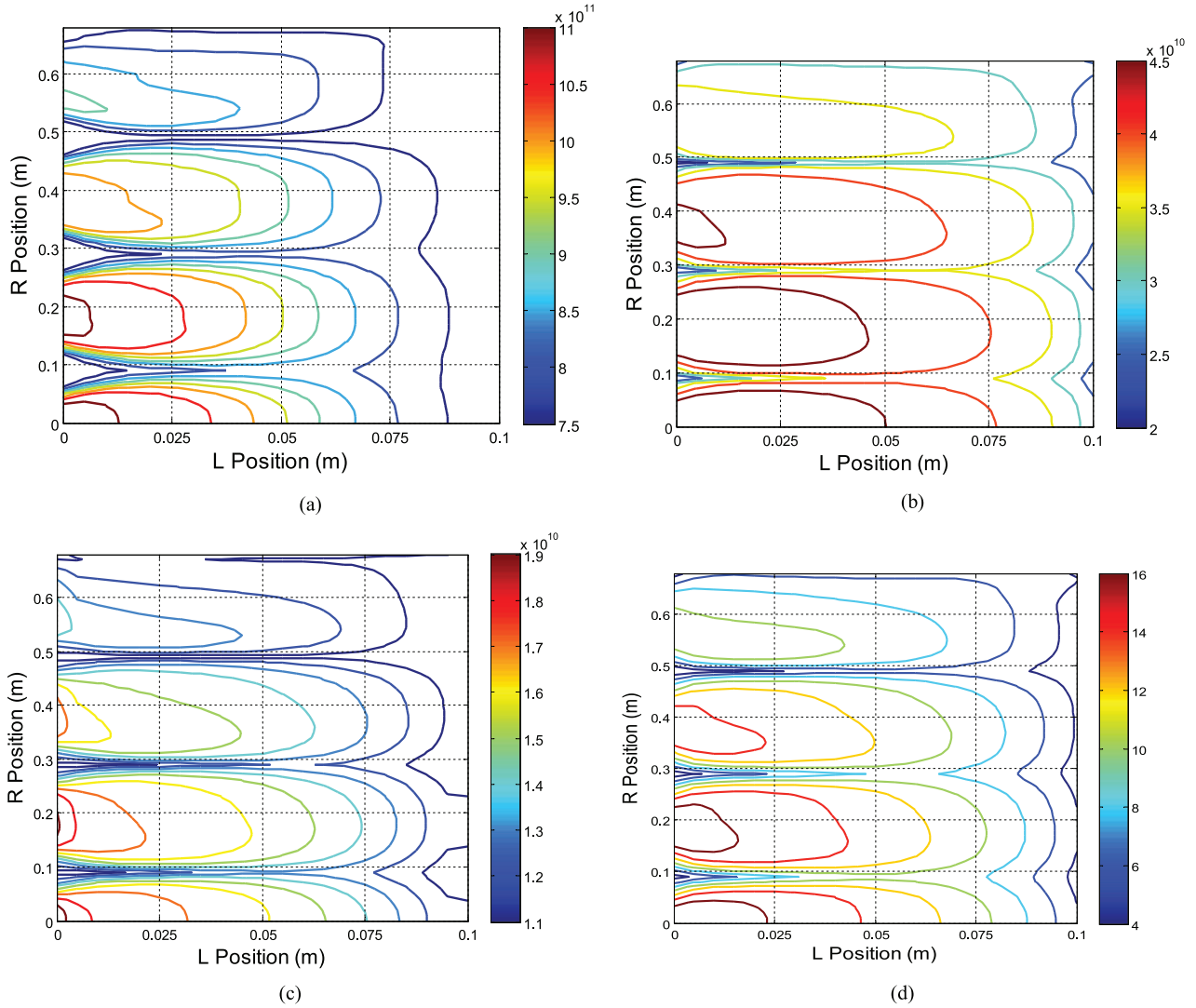


Figure 6. The calculated profiles as the mixing ratio of Ar/O₂/Cl₂ are 9:0.2:1 at 3 mTorr. (a) n_e , (b) n_{Cl} , (c) n_O , (d) T_e .

in [56]. In addition, ion densities have the similar property. But the increasing of the Cl and O densities is relatively less, it can be explained by the fact that at higher pressure recombination and charged exchange reactions ($O^- + O \rightarrow O_2^- + e$ and $Ar^+ + O \rightarrow O^+ + Ar$) become more important [6]. The electron temperature drops with the pressure increased, as the collision frequency does between electrons and neutrals. Therefore, electrons more easily transfer their energies to neutrals and decrease their kinetic energies. Electron temperatures in the magnetized region are about 4.2 eV, 4.4 eV and 4.4 eV, but near the cathode they are around 17.5 eV, 16.9 eV and 12.2 eV, respectively. The predicted variation of plasma density and electron temperature with pressure changed agree well with the published plasma modeling [5, 56].

4. Conclusions

In this work, we used a 2D hybrid model to describe the reaction chamber radius, the mixing ratio and pressure how to influence the Ar/O₂/Cl₂ discharge in the magnetized DC device. The

results showed that the main particles are Ar^{**}, Ar^{*}, e, Ar⁺, Cl, Cl₂⁺, O(³P) and O₂⁺ besides the background gases. The densities of oxygen species are slightly lower than those of Cl₂ species. The nonlinear variation of the plasma parameters shows that the important significance of design the magnetic field when the reaction chamber is expanded. If asymmetric magnetic poles are adopted which means that the magnetic field strength of the central magnetic pole is slightly weaker than those of the edge magnetic poles, it will improve the uniformity of the plasma density profiles at the film basal position. Even the mixing ratio of Ar/O₂/Cl₂ is varied from 9:0.2:1 to 9:1:0.2, the main particles are hardly changed. As the ratio of Cl₂ decreases, the concentrations of Cl₂⁺ ions and Cl atoms significantly decrease, but the concentrations of electrons, Ar intermediates and O atoms slightly increase. The main plasma particles are significantly influenced by the pressure. Electron density at 3 mTorr appears lower, but more inhomogeneous than that at 6 mTorr. The increasing of Cl and O densities is relatively less due to the recombination and charged exchange reactions become more important.

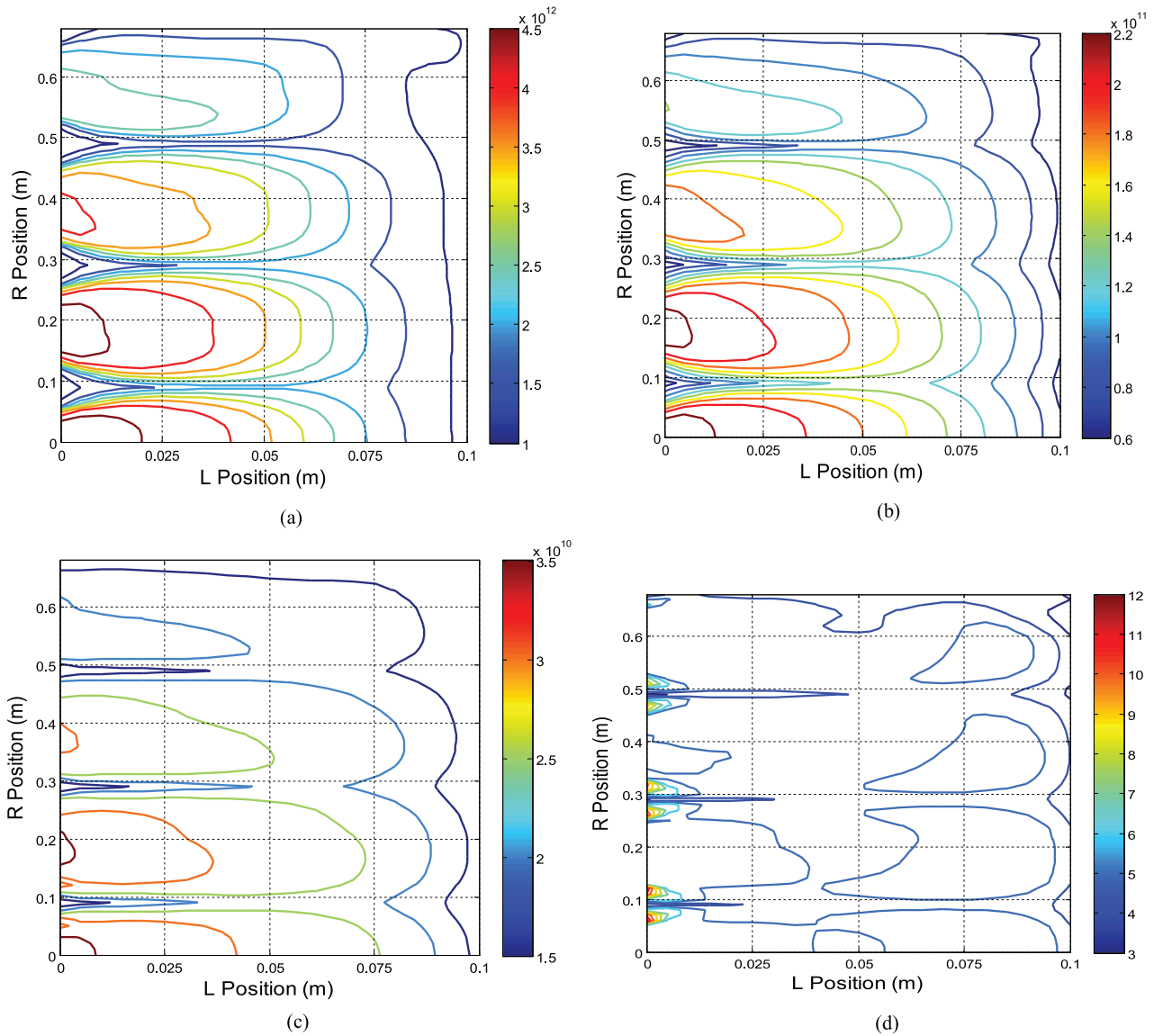


Figure 7. The calculated profiles as the mixing ratio of Ar/O₂/Cl₂ are 9:0.2:1 at 6 mTorr. (a) n_e , (b) n_{Cl} , (c) n_O , (d) T_e .

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